

Module 2 Project Rubric

TraceLens | Brownfield Challenge | AIE 500 - 10x Coding with AI

C1.1-C1.6 | Completes C1 when all six are Met.

Understand and repair the inherited TraceLens backend; verify AI-assisted changes against observable behavior.

Evidence to prepare

- Create docs/SYSTEM_MODEL.md: routes, request/storage flow, project/trace relationship, cost units, dependencies and context strategy.
- Close TL-1 through TL-6 against CONTRACT.md, retaining passing behavior and adding regression checks. Document your deletion policy and hand calculation.
- Keep docs/prompt-log.md with at least two substantive iterations and docs/ai-verification-note.md with one real, traceable AI error.
- Write accurate Google-style docstrings on changed/added functions and verify README setup from a clean copy.

MUST PASS: C1.4 - Every ticket is closed

A Not Yet returns this evidence group before its remaining criteria are reviewed.

Same competencies, independent project

Choose TraceLens **instead of PromptLab**. The videos demonstrate PromptLab; apply the methods independently. Both tracks require all 22 criteria to be Met, with no points, averaging or bonus credit.

No weekly capstone upload: keep one repository and authentic evidence for final review. Each module evidence group retains three attempts; returns do not consume one. See ASSESSMENT.md for revision, return and defense policy.

Version: TraceLens Criterion Set v1.1. Original project Modules 1-5 map to current Modules 2-6. The Syllabus governs policy; ALIGNMENT.md records clarifications.

Module 2 | Assessment criteria (1/3)

C1 - Codebase Comprehension & AI-Assisted Debugging

C1.1-C1.6 | Completes C1 when all six are Met.

Criterion	Where your evidence lives	What earns "Met"	Common mistakes
C1.1 - Your system model is accurate	docs/SYSTEM_MODEL.md	Correctly identifies every route the app exposes, the request-to-storage-and-back data flow, the project/trace relationship, where and how per-call cost is computed, how the storage layer works, and every external dependency. Nothing in the document contradicts the code.	- Describing what the AI said the code does rather than what it does. - Missing a module or dependency. - Listing file names without explaining behaviour.
C1.2 - You justify your context strategy	docs/SYSTEM_MODEL.md, context strategy section	States, for each stage of exploration, whether you used whole-repo or file-level context, with a reason based on the size or coupling of the code.	- Not stating which approach you used. - A reason that doesn't match your prompt log. - Whole-repo context throughout with no rationale.
C1.3 - Your prompt log shows real iteration	docs/prompt-log.md	At least two iterations where you narrowed context, added a constraint, or restructured a prompt because the output wasn't good enough - with your reasoning recorded.	- One successful prompt and nothing else. - Prompts listed with no outputs or reasoning. - Rewording without changing context or constraints.

Source: TraceLens Criterion Set v1.1 (rubrics/CRITERIA.md), adapted from supplied v1.0 and aligned to the current PromptLab course package. All criteria are Met / Not Yet.

Module 2 | Assessment criteria (2/3)

C1 - Codebase Comprehension & AI-Assisted Debugging

C1.1-C1.6 | Completes C1 when all six are Met.

Criterion	Where your evidence lives	What earns "Met"	Common mistakes
C1.4 - Every ticket is closed MUST PASS	Repository; pytest tests/ -v output	All 5 defects fixed - TL-1: 404 on a missing trace; TL-2: cost correct per million tokens, verified against a hand calculation; TL-3: pages contain at most limit items (short last pages and empty pages past the end are valid) and total is the full filtered count; TL-4: a project with no traces returns zeroed stats instead of a 500; TL-5: orphaned traces handled with your chosen strategy recorded - and TL-6: PATCH /projects/{id} implemented with partial updates, timestamp update and 404 handling. All provided tests pass and nothing that worked before is broken.	- A ticket still open. - Changing the test's expected cost instead of fixing the unit conversion. - Hiding the stats crash behind a try/except instead of handling the empty case. - Fixing the page size but leaving total wrong, or the reverse. - PATCH implemented but requiring all fields. - A fix that breaks something that worked before.

Contract clarification. A last page may be short and a page past the end empty. Record and test cascade, detach or reject for project deletion. All supplied diagnostics must pass after repairs; do not weaken them.

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Module 2 | Assessment criteria (3/3)

C1 - Codebase Comprehension & AI-Assisted Debugging

C1.1-C1.6 | Completes C1 when all six are Met.

Criterion	Where your evidence lives	What earns "Met"	Common mistakes
C1.5 - You caught the AI being wrong	docs/ai-verification-note.md, traceable to docs/prompt-log.md	At least one specific case of wrong AI output: what it produced, why it was wrong, how you spotted it, what you did instead. The case appears in your prompt log.	- General comments about AI limitations with no actual example. - An example that's just a syntax error your editor would have caught. - An example not traceable to the log.
C1.6 - Your documentation is accurate	Docstrings on functions you modified; README run instructions	Every function you modified or added has a Google-style docstring whose Args, Returns and Raises match the implementation. README run steps work on a clean clone.	- Docstrings listing parameters that don't exist or omitting ones that do. - README steps that fail on a fresh machine. - Docstrings restating the function name as a sentence.

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